



Synthesis of diphosphite ligands derived from glucopyranoside and their application in the Cu-catalyzed asymmetric 1,4-addition of organozinc to enones

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Dedicated to Professor Albert S. C. Chan on the occasion of his 60th birthday

ABSTRACT

Novel chiral diphosphite ligands derived from glucopyranoside and H₈-binaphthol were synthesized, and successfully employed in the Cu-catalyzed asymmetric 1,4-addition of organozinc reagents dimethylzinc, diethylzinc, and diphenylzinc to cyclic and acyclic enones with up to 96% ee. The stereochemically matched combination of D-glucopyranoside backbone and (R)-H₈-binaphthyl in the ligand 2,4-bis{[(R)-1,1'-H₈-binaphthyl-2,2'-diyl] phosphite}-phenyl 3,6-anhydro-β-D-glucopyranoside was essential for inducing high enantioselectivity. A significant dependence of stereoselectivity on the type of enones and the ring size of cyclic enones was observed. Moreover, the sense of the enantiodiscrimination of the products was mainly determined by the configuration of the H₈-binaphthyl moieties.

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1. Introduction

Carbohydrates are abundant and inexpensive naturally occurring, enantiomerically pure compounds that have considerably great structural variety. They consist of a carbon backbone and a high density of functional groups. Hence, carbohydrates have been extensively used as synthons for versatile ligands.¹ In recent years, chiral phosphites derived from carbohydrate skeletons (Fig. 1), particularly furanoside and pyranoside, are of particular interest and have been successfully applied in asymmetric catalytic reactions, such as hydrogenations,² hydroformylations,³ allylic alkylations,⁴ and 1,4-additions.⁵

The 1,4-addition of carbon nucleophiles to α,β-unsaturated compounds have been developed into one of the most powerful methods in organic chemistry for the formation of C–C bonds.⁶ The enantioselective 1,4-addition of organometallic reagents to cyclic and acyclic enones have been extensively investigated in recent decades, since the desired products are useful building blocks for the synthesis of enantiomerically enriched pharmaceuticals, agrochemicals, and natural products.^{6a} These results indicated that the transmetalation between organozinc reagents and transition

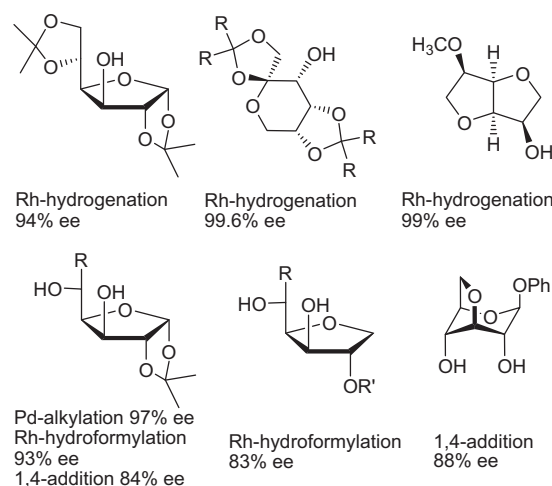


Figure 1. Selected carbohydrate backbone for the synthesis of chiral phosphite ligands.

metal complexes, such as Ni, Cu, Rh, and Ti salts in the presence of chiral ligands, generated more efficient catalysts in the reaction.

We are particularly interested in Cu-catalyzed reactions due to their low cost and wide range of catalytic activities, such as oxidations,^{7a} reductions,^{7b} addition reactions to multiple

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bonds^{7c,d}, and cross coupling reactions.^{7e,f} To date, several chiral phosphorus ligands, such as phosphoramidite,⁸ phosphite ligands,⁹ P, N-ligands,¹⁰ and others,¹¹ have been successfully employed in Cu-catalyzed asymmetric 1,4-additions of organozinc reagents to enones, and good results have been obtained. However, one of the limits revealed in these investigations is the dynamic equilibrium amongs several active species of organocopper compounds in the catalytic systems. If the more reactive cuprates accelerate the formation of racemic products, the loss of enantioselectivity is unavoidable. Hence, it is still highly important to carefully explore effective chiral phosphorus ligands, which can increase enantioselectivity and suppress the occurrence of the competing side reactions.

Diéguez et al. reported that diphosphite ligands **1a** and **1b** gave 81% ee (*R*) and 84% ee (*S*) in the Cu-catalyzed 1,4-addition of ZnEt₂ to 2-cyclohexenone **3a**, respectively.^{5a,b} Subsequently, Chan et al. reported diphosphite ligand **1c**, which was based on the pyranoside backbone of D-glucose, afforded 88% ee (*R*) in the above-mentioned reaction (see Fig. 2).^{9b} It was noted that heterodonor ligands, such as phosphine-phosphite, thioether-phosphite, and phosphoroamidite-phosphite, derived from a furanoid carbohydrate, had been tested separately in this reaction, and that enantioselectivities of no more than 63% ee were obtained.^{5a,b} It have been shown that diphosphite ligands derived from D-(+)-xylose or D-(+)-glucose are highly effective in the Cu-catalyzed 1,4-addition of ZnEt₂ to **3a**.

Based on previous results that the phosphite ligands were effective for 1,4-additions,⁹ while the C₁ symmetric furanoid and pyranoid ligands bearing 1,3-diphosphite moieties (Fig. 3) demonstrated good enantioselective control in the reaction,^{5a,b,9b} it was envisioned that changing the rigid 2,2'-dihydroxy-1,1'-binaphthyl (binaphthyl) substituents of the phosphites to the larger torsion dihedral angle structure of 2,2'-dihydroxy-5,5',6,6',7,7',8,8'-octahydro-1,1'-binaphthyl (H₈-binaphthyl) might result in a combination of these advantages (Fig. 3). Chan et al. reported that the steric tuning of the substituent H₈-binaphthyl on the biaryl-based diphosphite ligands was critical for achieving higher enantioselectivities in the Cu-catalyzed 1,4-addition of dialkylzincs to cyclic enones with up to 99% ee.^{9c} The drastic impact of the substituents on the ligands on enantioselectivity can be explained by the larger torsion dihedral angle and electronic property of H₈-binaphthyl in the Cu complexes, which constitute an integral part of the chiral environment around the Cu center. By taking advantage of the rigid tricycle structure and the remote stereocontrol capability of the substituents on

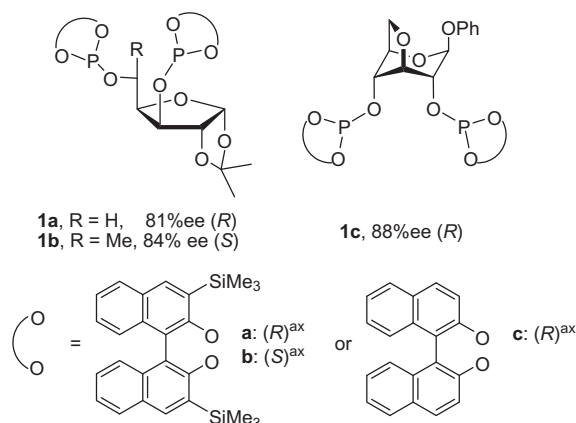


Figure 2. The representative examples of chiral diphosphite ligands prepared from D-(+)-xylose or D-(+)-glucose for the Cu-catalyzed asymmetric 1,4-addition of ZnEt₂ to 2-cyclohexenone.

the backbone of pyranoid ligands **1c** (Fig. 3), we decided to synthesize novel ligands **2a** and **2b** (Scheme 1) containing all of the elements above for the Cu-catalyzed 1,4-addition of ZnR₂ to cyclic and acyclic enones. Herein, we report the synthesis of novel diphosphite ligands, and their application in the Cu-catalyzed asymmetric 1,4-addition of ZnR₂ to cyclic and acyclic enones.

2. Results and discussion

2.1. Synthesis of diphosphite ligands 2a and 2b

As shown in Scheme 1, diphosphite ligands **2a** and **2b** were easily synthesized in one step from the corresponding phenyl 3,6-anhydro-β-D-glucopyranoside **5** and 5,5',6,6',7,7',8,8'-octahydro-1,1'-binaphthyl-2,2'-diyl-chlorophosphite **6**, which were conveniently obtained according to literature procedures.^{9b,12} Ligands **2a** and **2b** were purified on a silica gel column under a nitrogen atmosphere with moderate to good yields.

2.2. Asymmetric 1,4-conjugate addition of ZnR₂ to enones

Ligands **2a** and **2b** were tested in the Cu-catalyzed asymmetric 1,4-addition of ZnEt₂ to 2-cyclohexenone **3a**, and the results are

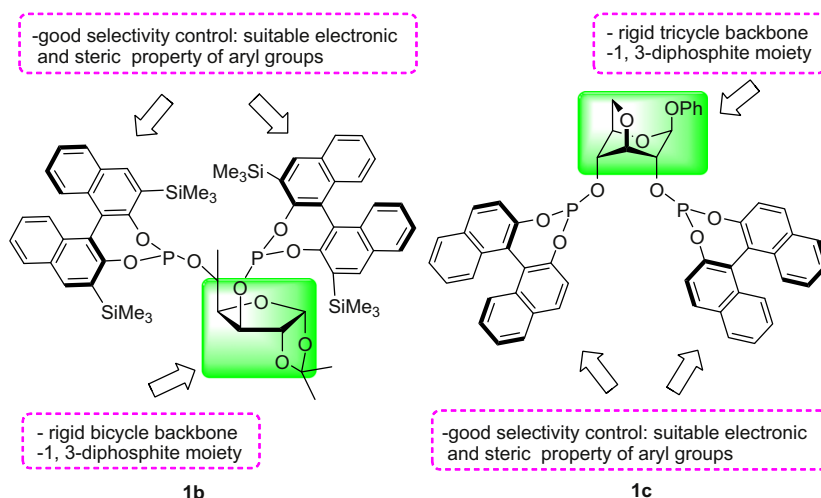
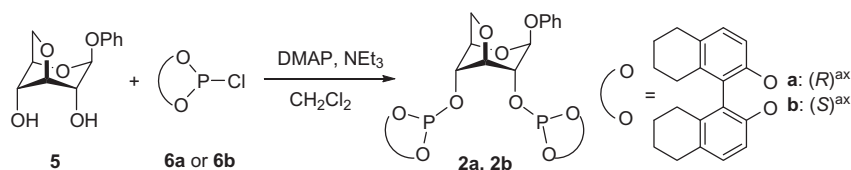


Figure 3. C₁ Symmetric structural features of furanoid ligand **1b** and pyranoid ligand **1c**.



Scheme 1. Synthesis of ligands **2a** and **2b** derived from glucopyranoside.

summarized in Table 1. In the first set of experiments, with $\text{Cu}(\text{OTf})_2$ as the copper precursor in THF, ligand **2a** afforded 77% yield and 82% ee (*R*) (Table 1, entry 1). Ligand **2b**, which was based on the same pyranoside backbone as ligand **2a** and the opposite configuration of the H_8 -binaphthol moieties, gave 71% yield and 15% ee (*S*) under the same reaction conditions (Table 1, entry 2). Alexakis and Feringa proposed that the catalytically active species was a Cu(I) complex,^{8d,10d} therefore, $(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$ was selected as the catalyst precursor instead of $\text{Cu}(\text{OTf})_2$ in the next entries. Up to 98% yield and 93% ee were reached in the presence of catalyst **2a**- $(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$ (Table 1, entry 3).

The influence of solvent and temperature were studied using a ligand **2a**- $(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$ catalytic system and **3a** as substrate. Enantioselectivities 75% ee (*R*) and 61% ee (*R*) were obtained in toluene and CH_2Cl_2 , respectively (Table 1, entries 5 and 6). The enantiomeric excess of 3-ethylcyclohexanone **4a** in Et_2O was equal to that in toluene, and a higher yield was obtained in Et_2O (Table 1, entries 4 and 5). It should be noted that THF as a solvent was beneficial for obtaining higher ee values of 93% (*R*) (Table 1, entry 3). When the temperature was decreased to -78°C , the reaction became sluggish and 73% ee (*R*) was obtained (Table 1, entry 10). A similar temperature effect was previously observed by Chan in the asymmetric 1,4-addition of ZnEt_2 to cyclic enones.^{9e} The best enantioselectivity was obtained at 0°C in our cases (Table 1, entry 3).

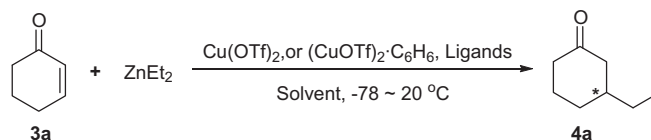
Under the aforementioned reaction conditions, we next examined the 1,4-addition of ZnEt_2 to 2-cyclopentenone **3b** and 2-cycloheptenone **3c** in the presence of copper(I) complexes of ligands **2a** and **2b**, and the results are shown in Table 2. The yield and enantioselectivity showed a significant dependence on the ring size of the substrates. Results of 90% ee (*R*) and 48% yield for product 3-ethylcyclopentanone **4b**, and 43% ee (*R*) and 63% yield for product 3-ethylcycloheptanone **4c** were obtained (Table 2, entries 1 and 2). It was found that the use of 2 equiv of ligand **2a** to copper salt gave the better results, although significant changes in the catalytic

activity or enantioselectivity were not observed in these cases (Table 1, entry 3, and Table 2, entries 1, 3–6).

The catalyst in situ prepared from $(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$ and ligand **2b** gave much lower catalytic activity and enantioselectivity (Table 2, entries 7 and 8). The results indicated that the matching combination of stereogenic centers of the glucopyranoside backbone and H_8 -binaphthyl moieties of ligand **2a** was fundamental to obtain higher catalytic activity and enantioselectivity. Under the optimal conditions, the performance of ligand **1c** was investigated, affording 71% ee (*R*) for **3b** and 35% ee (*R*) for **3c**, respectively. The results clearly show that ligand **2a** induced better enantioselectivity than its parent ligand **1c** (Table 2, entries 1, 2, 9, and 10). This observation suggests that the stereocontrol capability of ligand **2a** could be attributed to the larger torsion dihedral angle of H_8 -binaphthol.¹³

The Cu-catalyzed asymmetric 1,4-additions of other organozinc reagents, such as ZnMe_2 and ZnPh_2 , to 2-cyclopentenone were also assessed. Using ZnMe_2 as an organozinc reagent, the product 3-methylcyclohexanone **4d** was produced with moderate enantioselectivity (Table 2, entry 11). Under the optimized reaction conditions, the product 3-phenylcyclohexanone **4e** with 96% ee (*R*) was obtained (Table 2, entry 12).¹⁴ In 2004, Feringa reported that the highest enantioselectivity (94% ee) of **4e** was obtained in the Cu-catalyzed 1,4-addition of ZnPh_2 using a monodentate phosphoramidite ligand.^{14c} Next, the 1,4-additions of ZnEt_2 to aryl acyclic enones were examined (Scheme 2). However, low enantioselectivities were observed: 25% ee (*R*) for 1,3-diphenyl-2-propenone **3f**, 31% ee (*R*) for 3-(4-chlorophenyl)-1-phenyl-2-propenone **3g**, and 34% ee (*R*) for 3-(4-methoxyphenyl)-1-phenyl-2-propenone **3h** (Scheme 2). The results indicated that the newly developed phosphite ligand **2a** is inferior to these acyclic enones. It was found that the sense of the enantiodiscrimination of the products was mainly controlled by the configuration of the H_8 -binaphthyl moieties.

Table 1
Cu-catalyzed asymmetric 1,4-addition of Et_2Zn to 2-cyclohexenone^a

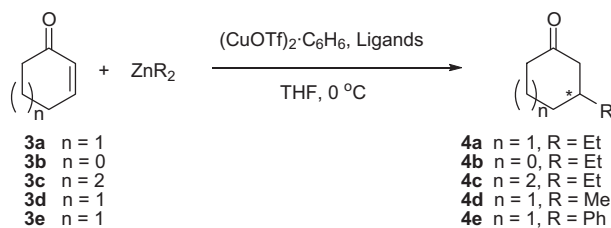


Entry	Cu precursor	Ligand	Solvent	<i>T</i> (°C)	<i>t</i> (h)	Conv. ^b (%)	Yield ^b (%)	%ee. (Conf.) ^c
1	$\text{Cu}(\text{OTf})_2$	2a	THF	0	4	>99	77	82 (<i>R</i>)
2	$\text{Cu}(\text{OTf})_2$	2b	THF	0	4	>99	71	15 (<i>S</i>)
3	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	THF	0	4	>99	98	93 (<i>R</i>)
4	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	Ether	0	4	>99	87	75 (<i>R</i>)
5	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	Toluene	0	4	>99	80	75 (<i>R</i>)
6	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	CH_2Cl_2	0	4	>99	82	61 (<i>R</i>)
7	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	THF	20	4	>99	90	91 (<i>R</i>)
8	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	THF	-20	15	>99	94	81 (<i>R</i>)
9	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	THF	-30	15	>99	90	78 (<i>R</i>)
10	$(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$	2a	THF	-78	15	21	12	73 (<i>R</i>)

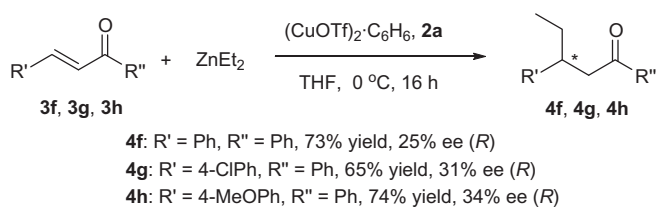
^a Reaction conditions: $\text{Cu}(\text{OTf})_2$ (0.01 mmol) or $(\text{CuOTf})_2 \cdot \text{C}_6\text{H}_6$ (0.005 mmol), ligand (0.02 mmol), ZnEt_2 (1.0 M in hexane, 1.2 mmol), **3a** (0.5 mmol), solvent (4 mL).

^b The data on conversion and yield were determined by GC using dodecane as an internal standard with a SE-30 column (30 m × 0.32 mm I.D.).

^c The enantiomeric excess of **4a** was determined by GC equipped with a Chiraldex A-TA column (50 m × 0.25 mm I.D.). The absolute configuration of **4a** was determined by comparison with an authentic sample.

Table 2Cu-catalyzed asymmetric 1,4-addition of ZnR₂ to cyclic enones^a

Entry	Ligand	Ligand/Cu	Enone	Product	Conv. ^b (%)	Yield ^b (%)	%ee (Conf.) ^b
1	2a	2	3b	4b	99	48	90 (R)
2 ^c	2a	2	3c	4c	81	63	43 (R)
3	2a	3	3a	4a	>99	97	93 (R)
4	2a	3	3b	4b	98	25	83 (R)
5	2a	1	3a	4a	97	82	93 (R)
6	2a	1	3b	4b	98	23	88 (R)
7	2b	2	3b	4b	98	8	23 (S)
8 ^c	2b	2	3c	4c	55	16	19 (S)
9	1c	2	3b	4b	92	36	71 (R)
10 ^c	1c	2	3c	4c	56	30	35 (R)
11 ^{c,d}	2a	2	3d	4d	25	18	65 (R)
12 ^e	2a	2	3e	4e	64	34	96 (R)

^a Reaction conditions: (CuOTf)₂·C₆H₆ (0.005 mmol), ligand (0.01–0.04 mmol), ZnEt₂ (1.0 M in hexane, 1.2 mmol), **3** (0.5 mmol), THF (4 mL), 4 h, 0 °C.^b The data on conversion, yield, enantiomeric excess, and the absolute configuration of the chiral product were determined using the same conditions as noted in Table 1.^c The enantiomeric excess was determined by GC equipped with a CP-Chirasil-Dex CB (25 m × 0.25 mm I.D.).^d ZnMe₂ (1.2 M in toluene, 1.2 mmol), 24 h.^e (CuOTf)₂·C₆H₆ (0.0025 mmol), **2a** (0.01 mmol), ZnPh₂ (1.2 M in toluene, 0.6 mmol), 20 h, isolated yield, the ee of **4e** was determined by HPLC (Daicel Chiralcel OD-H, Hexane/*i*-PrOH = 99.2/0.8, 0.5 mL/min at 20 °C, detected at 209 nm).**Scheme 2.** Cu-catalyzed 1,4-conjugate addition of ZnEt₂ to acyclic enones.

3. Conclusion

In conclusion, we have developed a new class of chiral diphosphite ligands derived from glucopyranoside and H₈-binaphthol, which has been successfully applied to the Cu-catalyzed asymmetric 1,4-addition of organozinc to a range of cyclic and acyclic enones with up to 96% ee. Although the enantioselectivity depends on the ring size of the substrate, the copper precursor, as well as the stereogenic centers of glucopyranoside backbone and the H₈-binaphthyl moieties of the ligands, the sense of enantiodiscrimination is mainly controlled by the H₈-binaphthyl moieties of the ligands. We have shown an example of the asymmetric addition of organozinc to enones here, the application of the ligands to other reactions is currently ongoing in our laboratory.

4. Experimental section

4.1. General

NMR spectra were recorded on Bulker 300 MHz and Bulker 400 MHz spectrometer at rt. ¹H NMR and ¹³C NMR spectra are reported in parts per million with TMS (δ = 0.00 ppm) as an internal standard. ³¹P NMR spectra are reported in parts per million with 85% H₃PO₄ as an external reference. Proton chemical shifts (δ)

and coupling constants (*J*) were given in ppm and in Hertz, respectively. Spin multiplicities are given as s (singlet), d (doublet), t (triplet), and m (multiplet) as well as br (broad). High resolution mass spectra (HRMS) were performed on a Bulker microTOF-QII mass instrument. All the melting points were determined on an X-4 melting point apparatus and are uncorrected. Optical rotations were measured on a Perkin-Elmer 241 MC polarimeter at 20 °C. Reactions were monitored by thin layer chromatography (TLC, silica gel GF254 plates). Column chromatography purifications were carried out using silica gel (200–300 mesh). Toluene, ether, THF, and NEt₃ were distilled from sodium and benzophenone. CH₂Cl₂ was distilled over CaH₂ under nitrogen. Phenyl 3,6-anhydro-β-D-glucopyranoside **5**^{9b} and 2,2'-dihydroxy-5,5',6,6',7,7',8,8'-octahydro-1,1'-binaphthol (H₈-binaphthol)¹² were synthesized according to literature procedures. All other chemicals were obtained commercially and used as received without further purification.

4.2. Synthesis of diphosphite **2a** and **2b**

4.2.1. 2,4-Bis[[(*R*)-1,1'-H₈-binaphthyl-2,2'-diyl] phosphite]-phenyl 3,6-anhydro-β-D-glucopyranoside **2a**

To a 100 mL Schlenk flask equipped with a condenser were charged (*R*)-H₈-binaphthol (1.77 g) in 20 mL of toluene and 10 mL of PCl₃. The mixture was stirred at 110 °C for 22 h under a nitrogen atmosphere. After removal of the excess PCl₃ and toluene, the residue was dissolved in toluene (20 mL). The solution was transferred to another Schlenk flask, and toluene was distilled in vacuo to obtain (*R*)-1,1'-H₈-binaphthyl-2,2'-diyl-chlorophosphine **6a** as a white powder, which was used directly in the following step without further purification. To a stirred solution of compound **5** (36 mg, 0.15 mmol), **6a** (215 mg, 0.6 mmol), and DMAP (4 mg, 0.033 mmol) in CH₂Cl₂ (10 mL), slowly added anhydrous NEt₃ (0.084 mL) at –15 °C under nitrogen for 0.5 h. The reaction mixture was then stirred at rt for 1 h. Next, CH₂Cl₂ was distilled off under reduced pressure, and then toluene (20 mL) was added. The solid was removed

by filtration through a pad of silica gel and the solvent was removed in vacuo. The residue was purified by flash chromatography (toluene, R_f = 0.63) and furnished ligand **2a** as a white foamy solid (90 mg, 68% yield), mp 131–132 °C. $[\alpha]_D^{20}$ = –255 (c 0.4, CH₂Cl₂). ³¹P NMR (121 MHz, DMSO-*d*₆) δ 139.42 (d, J = 6.1 Hz, 1P), 130.60 (d, J = 6.1 Hz, 1P). ¹H NMR (400 MHz, DMSO-*d*₆) δ 7.33 (t, J = 8.0 Hz, 2H), 6.94–7.27 (m, 9H), 6.89 (d, J = 8.0 Hz, 1H), 6.67 (d, J = 8.0 Hz, 1H), 5.48 (s, 1H), 4.54 (m, 1H), 4.35 (t, J = 2.8 Hz, 1H), 4.25 (m, 1H), 4.21 (d, J = 11.2 Hz, 1H), 3.89 (d, J = 10.0 Hz, 1H), 3.78 (dd, J = 10.0, 2.8 Hz, 1H), 2.30–2.83 (m, 12H), 2.08–2.16 (m, 4H), 1.66–1.72 (m, 12H), 1.43–1.51 (m, 4H). ¹³C NMR (100 MHz, DMSO-*d*₆) δ 155.9, 145.5, 145.4, 137.9, 137.2, 137.0, 134.8, 134.7, 133.8, 133.6, 129.6, 129.5, 129.4, 129.2, 128.6, 128.5, 128.4, 127.1, 126.6, 121.8, 118.8, 118.7, 118.2, 96.2, 73.9, 73.0, 71.5, 70.1, 69.1, 69.0, 28.4, 28.3, 27.2, 22.0, 21.9, 21.8. HRMS (ESI) calcd for: C₅₂H₅₃O₉P₂ (M+H)⁺ 883.3159, found: 883.3148, 1.2 ppm.

4.2.2. 2,4-Bis[(*S*)-1,1'-H₈-binaphthyl-2,2'-diyl] phosphite}-phenyl 3,6-anhydro- β -D-glucopyranoside **2b**

(*S*)-1,1'-H₈-Binaphthyl-2,2'-diyl-chlorophosphine **6b** was synthesized by the same procedure as that for **6a** and was used directly in the following step without further purification. Treatment of compound **5** (72 mg, 0.3 mmol), **6b** (430 mg, 1.2 mmol) and DMAP (8 mg, 0.066 mmol) as described for ligand **2a** afforded ligand **2b**, which was purified by flash chromatography (toluene, R_f = 0.45) to produce a white foamy solid (120 mg, 45% yield), mp 136 °C; $[\alpha]_D^{20}$ = +230 (c 0.4, CH₂Cl₂); ³¹P NMR (121 MHz, DMSO-*d*₆) δ 136.75 (s, 1P), 136.30 (s, 1P). ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.31 (t, J = 7.8 Hz, 2H), 7.25 (t, J = 7.5 Hz, 2H), 6.90–7.07 (m, 7H), 6.83 (d, J = 8.1 Hz, 1H), 6.73 (d, J = 7.8 Hz, 1H), 5.46 (d, J = 1.2 Hz, 1H), 4.36 (s, 2H), 4.20–4.24 (d, 2H), 3.87 (d, J = 10.2 Hz, 1H), 3.68 (d, J = 8.1 Hz, 1H), 2.56–2.76 (m, 12H), 2.06–2.17 (m, 4H), 1.63–1.71 (m, 12H), 1.35–1.51 (m, 4H). ¹³C NMR (75 MHz, DMSO-*d*₆) δ 156.2, 145.4, 137.9, 136.9, 134.7, 133.7, 129.5, 129.2, 129.1, 128.6, 128.5, 127.1, 126.9, 122.1, 118.8, 118.2, 116.3, 97.0, 75.0, 74.9, 72.7, 72.1, 70.3, 69.8, 28.4, 27.2, 27.1, 21.9, 21.8. HRMS (ESI) calcd for: C₅₂H₅₃O₉P₂ (M+H)⁺ 883.3159, found: 883.3140, 2.2 ppm.

4.3. Representative procedure for the 1,4-addition of diethylzinc to 2-cyclohexenone **3a**

A solution of (CuOTf)₂·C₆H₆ (0.005 mmol, 2.5 mg) and ligand **2a** (0.02 mmol, 17.7 mg) in THF (4 mL) was stirred for 1 h at rt under nitrogen. After the solution was cooled to 0 °C, 2-cyclohexenone (0.5 mmol, 0.048 mL) was added and the solution was stirred for 10 min at 0 °C. Then ZnEt₂ (1.2 mmol, 1.2 mL of 1.0 M solution in hexane) was added dropwise using a syringe within 2 min. After 4 h, the reaction was quenched by H₂O (2 mL) and 2 M HCl (2 mL), and extracted with ethyl acetate (5 mL $\times$ 3). The combined organic layer was washed with saturated NaHCO₃ solution, brine, and then dried over anhydrous Na₂SO₄, filtered and concentrated in vacuo to obtain the crude product. The conversion and the yield were determined by GC equipped with a SE-30 column (30 m $\times$ 0.32 mm I.D.) using dodecane as an internal standard. The enantiomeric excess was determined by GC analysis with a Chiraldex A-TA column (50 m $\times$ 0.25 mm I.D.). The absolute configuration was determined by comparison with authentic samples. The crude product was purified by column chromatography on silica gel (200–300 mesh) (Petroleum ether/EtOAc, 20/1) and the product 3-ethylcyclohexanone **4a** was obtained as a colorless oil. The characterization data and analytic conditions of the 1,4-adducts are as follows.

4.3.1. 3-Ethylcyclohexanone **4a**¹⁵

¹H NMR (400 MHz, CDCl₃) δ 2.15–2.38 (m, 3H), 1.82–2.01 (m, 3H), 1.52–1.69 (m, 2H), 1.19–1.34 (m, 3H), 0.84 (t, J = 7.6 Hz, 3H).

¹³C NMR (100 MHz, CDCl₃) δ 211.2, 46.9, 40.5, 39.8, 29.9, 28.3, 24.3, 10.2. Chiraldex A-TA column (50 m $\times$ 0.25 mm I.D.) at 120 °C constant. t_R = 11.52 min for enantiomer (*R*), and t_R = 11.79 min for enantiomer (*S*).

4.3.2. 3-Ethylcyclopentanone **4b**¹⁵

¹H NMR (400 MHz, CDCl₃) δ 2.23–2.38 (m, 2H), 2.03–2.16 (m, 3H), 1.73–1.80 (m, 1H), 1.40–1.50 (m, 3H), 0.92 (t, J = 7.6 Hz, 3H). ¹³C NMR (100 MHz, CDCl₃) δ 219.8, 44.9, 38.9, 38.4, 29.1, 28.4, 12.1. Chiraldex A-TA column (50 m $\times$ 0.25 mm I.D.) at 120 °C constant. t_R = 10.58 min for enantiomer (*R*), and t_R = 10.19 min for enantiomer (*S*).

4.3.3. 3-Ethylcycloheptanone **4c**^{14d}

¹H NMR (400 MHz, CDCl₃) δ 2.36–2.49 (m, 4H), 1.85–1.94 (m, 3H), 1.54–1.67 (m, 2H), 1.22–1.45 (m, 4H), 0.90 (t, J = 7.6 Hz, 3H). ¹³C NMR (100 MHz, CDCl₃) δ 214.9, 49.6, 43.9, 37.7, 36.5, 30.1, 28.6, 24.4, 11.4. CP-Chirasil-Dex CB (25 m $\times$ 0.25 mm I.D.) at 120 °C constant. t_R = 13.70 min for enantiomer (*R*), and t_R = 13.31 min for enantiomer (*S*).

4.3.4. 3-Methylcyclohexanone **4d**^{14d}

¹H NMR (300 MHz, CDCl₃) δ 2.18–2.41 (m, 3H), 2.01–2.09 (m, 2H), 1.83–1.95 (m, 2H), 1.59–1.74 (m, 1H), 1.24–1.40 (m, 1H), 1.02 (d, J = 6.3 Hz, 3H). ¹³C NMR (75 MHz, CDCl₃) δ 212.0, 50.0, 41.1, 34.2, 33.3, 25.3, 22.1. CP-Chirasil-Dex CB (25 m $\times$ 0.25 mm I.D.) at 70 °C constant. t_R = 37.36 min for enantiomer (*R*), and t_R = 39.01 min for enantiomer (*S*).

4.3.5. 3-Phenylcyclohexanone **4e**¹⁵

¹H NMR (400 MHz, CDCl₃) δ 7.31–7.35 (m, 2H), 7.21–7.25 (m, 3H), 2.97–3.05 (m, 1H), 2.34–2.59 (m, 4H), 2.06–2.18 (m, 2H), 1.79–1.91 (m, 2H). ¹³C NMR (100 MHz, CDCl₃) δ 211.2, 144.4, 128.7, 126.7, 126.6, 49.0, 44.8, 41.2, 32.8, 25.6. HPLC (Daicel Chiralcel OD-H 25 cm $\times$ 4.6 mm I.D., Hexane/*i*-PrOH = 99.2/0.8, flow rate = 0.5 mL/min at 20 °C, detected at 209 nm): t_R = 36.98 min for enantiomer (*R*), and t_R = 34.29 min for enantiomer (*S*).

4.3.6. 1,3-Diphenyl-1-pentanone **4f**^{14d}

¹H NMR (400 MHz, CDCl₃) δ 7.90 (d, J = 7.2 Hz, 2H), 7.53 (d, J = 7.2 Hz, 1H), 7.53 (d, J = 7.6 Hz, 2H), 7.16–7.30 (m, 5H), 3.22–3.30 (m, 3H), 1.76–1.84 (m, 1H), 1.60–1.70 (m, 1H), 0.80 (t, J = 7.6 Hz, 3H). ¹³C NMR (100 MHz, CDCl₃) δ 199.2, 144.7, 137.3, 132.9, 128.5, 128.4, 128.0, 127.6, 126.3, 45.6, 43.0, 29.2, 12.1. HPLC (Daicel Chiralcel AD-H 25 cm $\times$ 4.6 mm I.D., Hexane/*i*-PrOH = 95/5, flow rate = 0.5 mL/min at 25 °C, detected at 254 nm): t_R = 16.75 min for enantiomer (*R*), and t_R = 14.90 min for enantiomer (*S*).

4.3.7. 3-(4-Chlorophenyl)-1-phenylpentanone **4g**^{10e}

¹H NMR (400 MHz, CDCl₃) δ 7.89 (d, J = 7.2 Hz, 2H), 7.54 (d, J = 7.6 Hz, 1H), 7.43 (d, J = 7.6 Hz, 2H), 7.24 (d, J = 8.8 Hz, 2H), 7.16 (d, J = 8.4 Hz, 2H), 3.24 (s, 3H), 1.74–1.81 (m, 1H), 1.57–1.64 (m, 1H), 0.80 (t, J = 7.6 Hz, 3H). ¹³C NMR (100 MHz, CDCl₃) δ 198.8, 143.1, 137.1, 133.0, 131.9, 129.0, 128.6, 128.5, 128.0, 45.4, 42.4, 29.2, 12.0. HPLC (Daicel Chiralcel AD-H 25 cm $\times$ 4.6 mm I.D., Hexane/*i*-PrOH = 95/5, flow rate = 0.5 mL/min at 25 °C, detected at 254 nm): t_R = 18.49 min for enantiomer (*R*), and t_R = 14.79 min for enantiomer (*S*).

4.3.8. 3-(4-Methoxyphenyl)-1-phenylpentanone **4h**^{14d}

¹H NMR (400 MHz, CDCl₃) δ 7.89 (d, J = 7.2 Hz, 2H), 7.52 (d, J = 7.2 Hz, 1H), 7.42 (d, J = 7.6 Hz, 2H), 7.13 (d, J = 8.8 Hz, 2H), 6.82 (d, J = 8.4 Hz, 2H), 3.77 (s, 3H), 3.18–3.24 (m, 3H), 1.73–1.79 (m, 1H), 1.58–1.64 (m, 1H), 0.80 (t, J = 7.6 Hz, 3H). ¹³C NMR (100 MHz, CDCl₃) δ 199.4, 157.9, 137.3, 136.7, 132.9, 128.5,

128.4, 128.1, 113.8, 55.2, 45.8, 42.3, 29.3, 12.1. HPLC (Daicel Chiralcel AD-H 25 cm $\times$ 4.6 mm I.D., Hexane/*i*-PrOH = 95/5, flow rate = 0.5 mL/min at 25 °C, detected at 254 nm): t_R = 23.76 min for enantiomer (*R*), and t_R = 16.93 min for enantiomer (*S*).

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